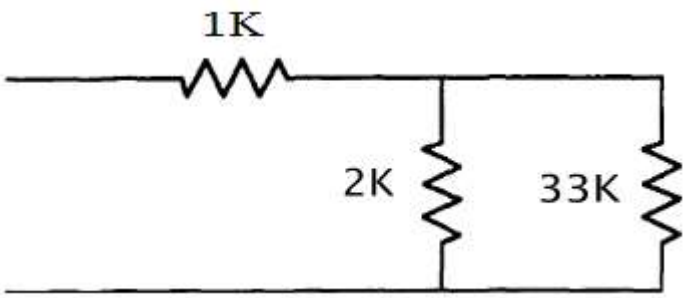
ECE 1101L: Lab 4 - Series & Parallel DC Circuits 3

Introduction: In this lab we used serial resistor and parallel resistors to solve our currents and voltages. Voltage division, current division, KCL, and KVL were all the rules we used for our theoretical data. For the first circuit our power supply was 10.00V so our calculations were done with this voltage. In the second circuit we had a voltage of 15.00V so the same was done. Lastly, our third circuit was done at 10.00V so we used that voltage for our calculations. Lastly, the turn of the potentiometer might not have been turned exactly to what was stated so there is some uncertainty there.

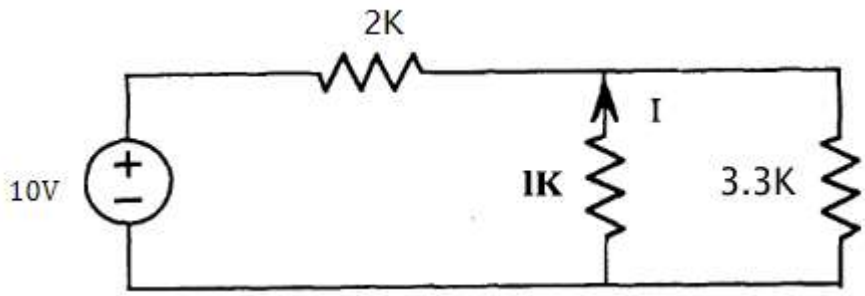
Pre-Lab:

1. Find the equivalent resistance of



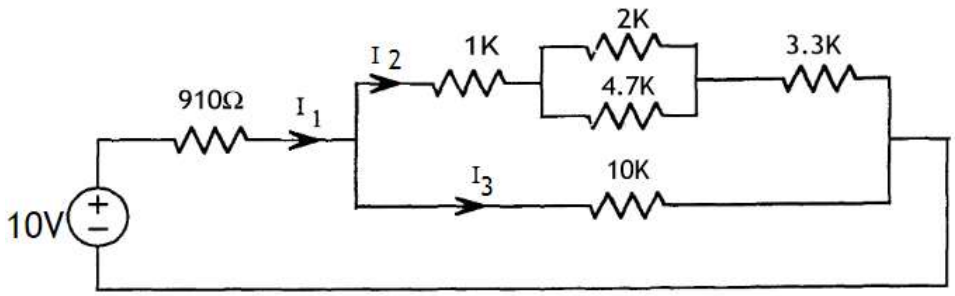
The equivalent resistance is 2.89kΩ

2. Find I



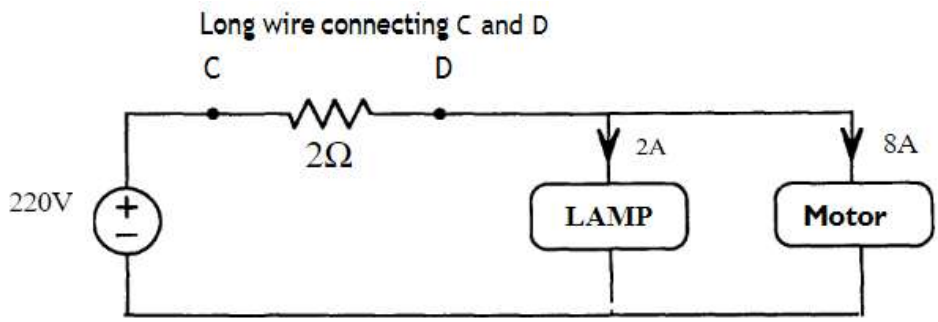
The current I is –2.77mA

3. Given



a. I1 = 1.1mA & I3 = 0.4mA
b. I1 – I3 = I2 => 1.1mA - 0.4mA = 0.7mA => I2 = 0.7mA

4. Given



a. Voltage drop is 20v
b. Voltage over lamp and motor is –200V because they are both parallel

Circuit 1

Voltage	Theory	Measured	Deviation
VA	10V	9.931V	0.69%
VB	5.997V	5.939V	0.97%
VC	5.997V	5.888V	1.81%
Total Resistance	2.499kΩ	2.513kΩ	-0.56%

Current	Theory	Measured	Deviation
R1	4.003mA	4.004mA	-0.02%
R2	2.726mA	2.725mA	0.04%
R3	1.276mA	1.253mA	0.02%

Explanation: First we got our theoretical values. We used the parallel resistor rule to add R2 and R3 and then added R2||3 with R1 to get total resistance. Then we calculated our theoretical voltages at points A, B, and C. After that we measured our total resistance. We calculated our theoretical current using

ohm's law, our voltage value, and our resistance values. Lastly, we measured our current values and did the deviation.

Circuit 2

Voltage	Theory	Measured	Deviation
VB	15V	14.962V	0.25%
VC	15V	14.958V	0.28%
VD	12.591V	12.590V	0.007%
VE	7.445V	7.438V	0.09%
Total Resistance	0.932kΩ	0.923kΩ	0.96%

Current	Theory	Measured	Deviation
Source	16.095mA	15.96mA	0.84%
R1	15mA	15.07mA	-0.47%
R2	1.0949mA	1.0994mA	-0.41%

Explanation: R2, R3, and R4 are all serial so they can be added together, then they are parallel with R1 so here we apply the parallel resistor rule. We calculated our total theoretical resistance and used this to be able to calculate theoretical voltages. We then calculated our theoretical current values. After this we measured total resistance, voltage at points B, C, D, and E, and current at Source, R1, and R2. With all our data we were able to calculate the deviation.

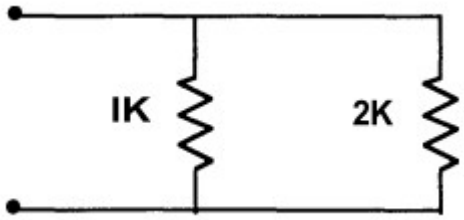
Circuit 3

Position	V_{WB} Open	VWB 47k	V_{WB} 4.7k	VWB 1k
Measure Fully CCW	18.217mV	18.209mV	18.140mV	8.777mV
Measure 1/4	1.5940V	1.8868V	1.6109V	0.76694V
Measure 1/2	4.5543V	4.3036V	2.8764V	1.2058V
Measure 3/4	7.4690V	7.4509V	4.5268V	1.9953V
Measure Fully CW	8.8889V	8.7321V	7.2156V	4.5594V

Explanation: In this circuit we used a 10k potentiometer. We measured the voltage of different resistors connected to the wiper of the potentiometer at different clockwise distances. We first measured all the resistance values at wiper fully counterclockwise. Then we repeated the measuring process for all the resistors at ¼ clockwise, ½ clockwise, ¾ clockwise, and fully clockwise.

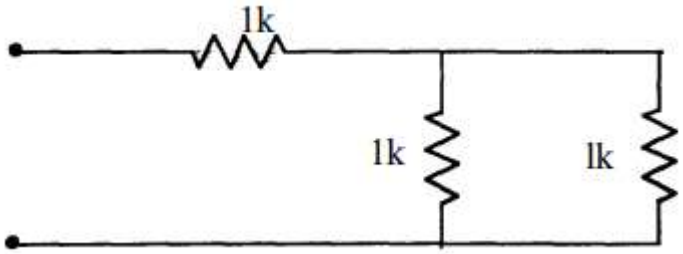
Post-Lab:

1. Suppose someone calculated that the equivalent resistance of the figure was 1.5k. How could you tell – without doing any calculations – that an error had been made.



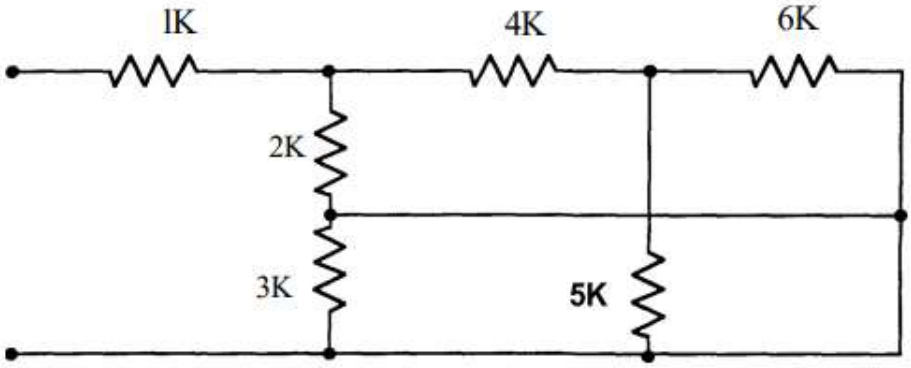
Without doing any calculations I would assume it's wrong because I know any way of multiplying these numbers and dividing them would not equate to 1.5kΩ.

2. Suppose someone calculated that the equivalent resistance of the figure was 900Ω. How could you tell – without doing any calculations – that an error had been made.



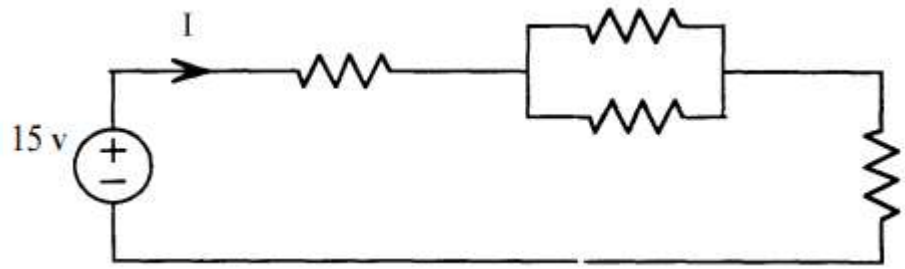
I can tell an error had been made because one the parallel resistors combine they are serial with the resistor at the top so the value of the equivalent resistance should at least be 1kΩ.

3. Find the equivalent resistance of the figure below. (Note that there are connections only where there are filled in circles)



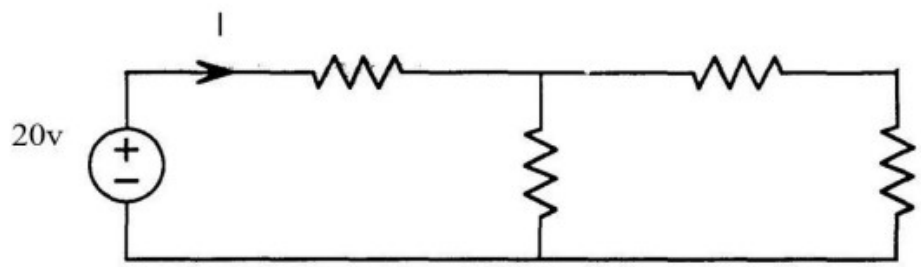
The equivalent resistance of the figure is 14.875kΩ

4. Suppose that in the circuit we measure I = 5mA. What is the equivalent resistance of the circuit?



The equivalent resistance of the circuit is 3kΩ

5. Suppose that in the circuit we measure I = 10mA. What will I be if we change the voltage supply to 25 volts.



The current will be 12.5mA if the voltage supply was changed to 25 volts.

Conclusion: In conclusion throughout this lab, we applied both parallel and serial resistor rules to be able to calculate our total resistance in all our circuits. These rules are super important because the total resistance helps us calculate the rest of our data. We also used a potentiometer and saw how the data differed at different rotations.